

## Questions

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## Q1 - 2024 (04 Apr Shift 1)

Let  $\alpha$  and  $\beta$  be the sum and the product of all the non-zero solutions of the equation  $(\bar{z})^2 + |z| = 0, z \in \mathbb{C}$ .

Then  $4(\alpha^2 + \beta^2)$  is equal to :

(1) 6

(2) 8

(3) 2

(4) 4

## Q2 - 2024 (04 Apr Shift 2)

The area (in sq. units) of the region  $S = \{z \in \mathbb{C} : |z - 1| \leq 2; (z + \bar{z}) + i(z - \bar{z}) \leq 2, \text{Im}(z) \geq 0\}$  is

(1)  $\frac{7\pi}{3}$

(2)  $\frac{7\pi}{4}$

(3)  $\frac{17\pi}{8}$

(4)  $\frac{3\pi}{2}$

## Q3 - 2024 (05 Apr Shift 1)

Consider the following two statements :

Statement I : For any two non-zero complex numbers  $z_1, z_2$ ,

$$(|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq 2(|z_1| + |z_2|), \text{ and}$$

Statement II : If  $x, y, z$  are three distinct complex numbers and  $a, b, c$  are three positive real numbers such that

$$\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}, \text{ then}$$

$$\frac{a^2}{y-z} + \frac{b^2}{z-x} + \frac{c^2}{x-y} = 1.$$

Between the above two statements,

(1) Statement I is correct but Statement II is incorrect.

(2) both Statement I and Statement II are correct.

(3) both Statement I and Statement II are incorrect.

(4) Statement I is incorrect but Statement II is correct.

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## Q4 - 2024 (05 Apr Shift 2)

Let  $S_1 = \{z \in \mathbb{C} : |z| \leq 5\}$ ,  $S_2 = \left\{z \in \mathbb{C} : \operatorname{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) \geq 0\right\}$  and  $S_3 = \{z \in \mathbb{C} : \operatorname{Re}(z) \geq 0\}$ . Then the area of the region  $S_1 \cap S_2 \cap S_3$  is :

(1)  $\frac{125\pi}{12}$

(2)  $\frac{125\pi}{4}$

(3)  $\frac{125\pi}{24}$

(4)  $\frac{125\pi}{6}$

## Q5 - 2024 (06 Apr Shift 2)

If  $z_1, z_2$  are two distinct complex number such that  $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right| = 2$ , then

(1)  $z_1$  lies on a circle of radius  $\frac{1}{2}$  and  $z_2$  lies on a circle of radius 1 .

(2) both  $z_1$  and  $z_2$  lie on the same circle. both  $z_1$  and  $z_2$  lie on the same circle.

(3) either  $z_1$  lies on a circle of radius  $\frac{1}{2}$  or  $z_2$  lies on a circle of radius 1 .

(4) either  $z_1$  lies on a circle of radius 1 or  $z_2$  lies on a circle of radius  $\frac{1}{2}$ .

## Q6 - 2024 (08 Apr Shift 1)

Let  $z$  be a complex number such that  $|z+2| = 1$  and  $\operatorname{Im}\left(\frac{z+1}{z+2}\right) = \frac{1}{5}$ . Then the value of  $|\operatorname{Re}(z+2)|$  is

(1)  $\frac{2\sqrt{6}}{5}$

(2)  $\frac{24}{5}$

(3)  $\frac{1+\sqrt{6}}{5}$

(4)  $\frac{\sqrt{6}}{5}$

## Q7 - 2024 (08 Apr Shift 2)

The sum of all possible values of  $\theta \in [-\pi, 2\pi]$ , for which  $\frac{1+i\cos\theta}{1-2i\cos\theta}$  is purely imaginary, is equal

(1)  $3\pi$

(2)  $2\pi$

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(3)  $5\pi$

(4)  $4\pi$

Q8 - 2024 (09 Apr Shift 1)

The sum of the square of the modulus of the elements in the set

$\{z = a + ib : a, b \in \mathbf{Z}, z \in \mathbf{C}, |z - 1| \leq 1, |z - 5| \leq |z - 5i|\}$  is \_\_\_\_\_

Q9 - 2024 (09 Apr Shift 2)

Let  $z$  be a complex number such that the real part of  $\frac{z-2i}{z+2i}$  is zero. Then, the maximum value of  $|z - (6 + 8i)|$  is equal to

(1) 12

(2) 10

(3) 8

(4)  $\infty$

Questions

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Answer Key

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Q1 (4) mathongo Q2 (4) mathongo Q3 (1) mathongo Q4 (1) mathongo

Q5 (4) mathongo Q6 (1) mathongo Q7 (1) mathongo Q8 (9) mathongo

Q9 (1) mathongo mathongo mathongo mathongo mathongo mathongo

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## Solutions

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Q1

$$z = x + iy$$

$$\bar{z} = x - iy$$

$$\bar{z}^2 = x^2 - y^2 - 2ixy$$

$$\Rightarrow x^2 - y^2 - 2ixy + \sqrt{x^2 + y^2} = 0$$

$$\Rightarrow x = 0 \quad \text{or} \quad -y^2 + |y| = 0$$

$$y = 0 \quad x^2 + |x| = 0$$

$$|y| = |y|^2$$

$$y = 0, \pm 1$$

$$\Rightarrow i, -i$$

are roots

$$\Rightarrow x = 0$$

$$\Rightarrow \alpha = i - i = 0$$

$$\beta = i(-i) = 1$$

$$4(0 + 1) = 4$$

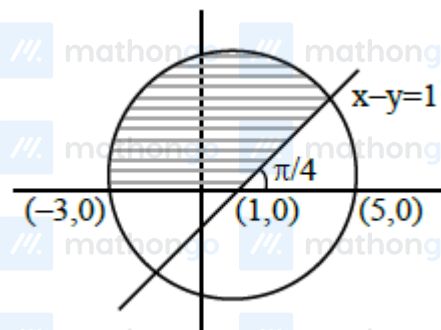
Q2

Put  $z = x + iy$ 

$$|z - 1| \leq 2 \Rightarrow (x - 1)^2 + y^2 \leq 4 \dots (i)$$

$$\Rightarrow x - y \leq 1 \dots (ii)$$

$$\text{Im}(z) \geq 0 \Rightarrow y \geq 0 \dots (iii)$$



Required area

= Area of semi-circle - area of sector A

$$\frac{1}{2}\pi(2)^2 - \frac{\pi}{2}$$

$$= \frac{3\pi}{2}$$

Q3

Statement I :

$$(|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right|$$

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## Solutions

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Since  $\left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq \left| \frac{z_1}{|z_1|} \right| + \left| \frac{z_2}{|z_2|} \right|$   
 $\left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq \frac{|z_1|}{|z_1|} + \frac{|z_2|}{|z_2|}$   
 $\left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \leq 2$   
 $(|z_1| + |z_2|) \left( \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right| \right) \leq 2(|z_1| + |z_2|)$   
 $\therefore$  statement I is correct

For Statement II :

$$\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$$

$$\frac{a^2}{|y-z|^2} = \frac{b^2}{|z-x|^2} = \frac{c^2}{|x-y|^2} = \lambda$$

$$a^2 = \lambda (|y-z|^2) = \lambda(y-z)(\bar{y}-\bar{z})$$

$$b^2 = \lambda(z-x)(\bar{z}-\bar{x}) \text{ and } c^2 = \lambda(x-y)(\bar{x}-\bar{y})$$

$$\frac{a^2}{y-z} + \frac{b^2}{z-x} + \frac{c^2}{x-y} = \lambda(\bar{y}-\bar{z} + \bar{z}-\bar{x} + \bar{x}-\bar{y}) = 0$$

Statement II is false

Q4

$$S_1 : x^2 + y^2 \leq 25 \dots (1)$$

$$S_2 : \operatorname{Im} \text{ of } \frac{z + (1 - \sqrt{3}i)}{(1 - \sqrt{3}i)} \geq 0$$

$$\operatorname{Im} \text{ of } \left( \frac{x + iy}{1 - \sqrt{3}i} + 1 \right) \geq 0$$

$$\operatorname{Im} \text{ of } \left( \frac{(x + iy)(1 + \sqrt{3}i)}{4} \right) \geq 0$$

$$\Rightarrow \sqrt{3}x + y \geq 0 \dots (2)$$

$$S_3 : x \geq 0 \dots (3)$$

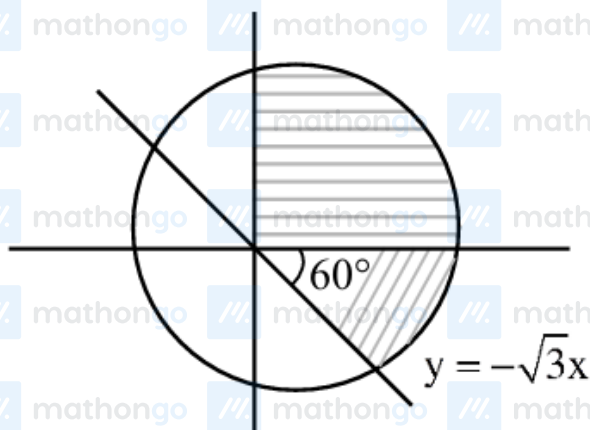
$$\text{Area} = \frac{5}{12} (\pi(5)^2)$$

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## Solutions

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Q5

$$\frac{z_1 - 2z_2}{\frac{1}{2} - z_1 \bar{z}_2} \times \frac{\bar{z}_1 - 2\bar{z}_2}{\frac{1}{2} - \bar{z}_1 z_2} = 4$$

$$|z_1|^2 2z_1 \bar{z}_2 - 2\bar{z}_1 z_2 + 4|z_2|^2$$

$$= 4 \left( \frac{1}{4} - \frac{\bar{z}_1 z_2}{2} - \frac{z_1 \bar{z}_2}{2} + |z_1|^2 |z_2|^2 \right)$$

$$z_1 \bar{z}_1 + 2z_2 \cdot 2\bar{z}_2 - z_1 \bar{z}_1 2z_2 2\bar{z}_2 - 1 = 0$$

$$(z, \bar{z}_1 - 1)(1 - 2z_2 \cdot 2\bar{z}_2) = 0$$

$$(|z_1|^2 - 1)(|2z_2|^2 - 1) = 0$$

Q6

$$|z + 2| = 1, \operatorname{Im}\left(\frac{z+1}{z+2}\right) = \frac{1}{5}$$

$$\text{Let } z + 2 = \cos \theta + i \sin \theta$$

$$\frac{1}{z + 2} = \cos \theta - i \sin \theta$$

$$\Rightarrow \frac{z + 1}{z + 2} = 1 - \frac{1}{z + 2} = 1 - (\cos \theta - i \sin \theta)$$

$$= (1 - \cos \theta) + i \sin \theta$$

$$\operatorname{Im}\left(\frac{z + 1}{z + 2}\right) = \sin \theta, \sin \theta = \frac{1}{5}$$

$$\cos \theta = \pm \sqrt{1 - \frac{1}{25}} = \pm \frac{2\sqrt{6}}{5}$$

$$|\operatorname{Re}(z + 2)| = \frac{2\sqrt{6}}{5}$$

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## Solutions

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Q7

$$Z = \frac{1 + i \cos \theta}{1 - 2i \cos \theta}$$

$$Z = -\bar{Z} \Rightarrow \frac{1 + i \cos \theta}{1 - 2i \cos \theta} = -\left( \frac{1 + i \cos \theta}{1 - 2i \cos \theta} \right)$$

$$(1 + i \cos \theta)(1 - 2i \cos \theta) = -(1 - 2i \cos \theta)(1 + i \cos \theta)$$

$$(1 + i \cos \theta)(1 + 2i \cos \theta) = -(1 - 2i \cos \theta)(1 - i \cos \theta)$$

$$1 + 3i \cos \theta - 2 \cos^2 \theta = -(1 - 3i \cos \theta - 2 \cos^2 \theta)$$

$$2 - 4 \cos^2 \theta = 0$$

$$\Rightarrow \cos^2 \theta = \frac{1}{2} \Rightarrow \theta = -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$\text{sum} = 3\pi$$

Q8

$$|z - 1| \leq 1$$

$$\Rightarrow |(x - 1) + iy| \leq 1$$

$$\Rightarrow \sqrt{(x - 1)^2 + y^2} \leq 1$$

$$\Rightarrow (x - 1)^2 + y^2 \leq 1 \dots (1)$$

$$\text{Also } |z - 5| \leq |z - 5i|$$

$$(x - 5)^2 + y^2 \leq x^2 + (y - 5)^2$$

$$-10x \leq -10y$$

$$\Rightarrow x \geq y \dots (2)$$

Solving (1) and (2)

$$\Rightarrow (x - 1)^2 + x^2 = 1$$

$$\Rightarrow 2x^2 - 2x = 0$$

$$\Rightarrow x(x - 1) = 0$$

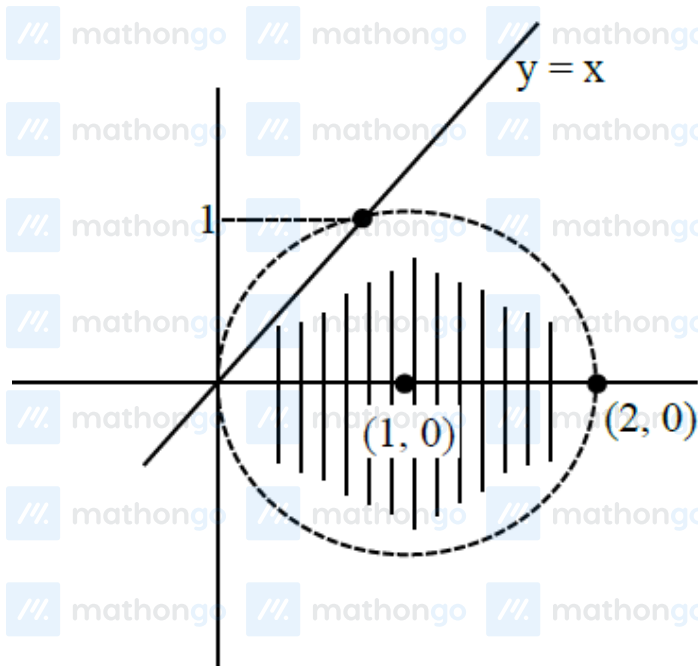
$$\Rightarrow x = 0 \text{ or } x = 1$$

$$y = 0 \text{ or } y = 1$$



## Solutions

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Given  $x, y \in \mathbb{I}$

Points  $(0, 0), (1, 0), (2, 0), (1, 1), (1, -1)$  to find

$$|z_1|^2 + |z_2|^2 + |z_3|^2 + |z_4|^2 + |z_5|^2 \\ = 0 + 1 + 4 + 1 + 1 + 1 + 1 = 9$$

Q9

$$\frac{z - 2i}{z + 2i} + \frac{\bar{z} + 2i}{\bar{z} - 2i} = 0$$

$$z\bar{z} - 2i\bar{z} - 2iz + 4(-1)$$

$$+ z\bar{z} + 2zi + 2\bar{z}i + 4(-1) = 0$$

$$\Rightarrow 2|z|^2 = 8 \Rightarrow |z| = 2$$

$$|z - (6 + 8i)|_{\text{maximum}} = 10 + 2 = 12$$

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